Egyptian History AI Agent

Project Plan: Bi-weekly Deliveries

Prepared for the six-member project team | English

1 Plan overview

This plan organizes the project into three deliveries, one every two weeks, ending with the final delivery on Thursday, 3 December 2026. Work weeks run Sunday to Thursday. The plan follows the priority order agreed with the product owner and does not assign roles to individual members; the team divides the work packages itself.

Item	Detail
Start date	Sunday, 1 November 2026
Final delivery	Thursday, 3 December 2026
Cadence	Delivery every two weeks (12 Nov, 26 Nov), then the final delivery (3 Dec)
Total duration	About 4.5 weeks: two full two-week sprints plus a short final week
Feature freeze	Thursday, 26 November 2026 (after Delivery 2)

2 Scope for this timeline

The full product (P0 to P6) is larger than the available time, so the scope is split into three groups.

Group	Priorities	Content
Must deliver	P0, P1, P2, P3	Trusted foundations, museum map and discovery, grounded Concierge and history, and the three historical characters (Khufu, Ahmose I, Cleopatra VII).
If time allows	P4	Self-guided tours with ordered stops and progress.
Deferred to roadmap	P5, P6	Accounts and saved state, and turn-based push-to-talk voice. Arabic and English remain required from the first delivery.

3 Delivery schedule

Delivery	Date	Focus	Evidence of completion
Delivery 1	Thu, 12 Nov 2026	Foundation and discovery (P0, P1)	Containers running; a visitor can search a place or artefact and see approved details.
Delivery 2	Thu, 26 Nov 2026	Grounded Concierge and history (P2)	Questions route correctly; every answer is grounded, scoped, and cited.

Delivery	Date	Focus	Evidence of completion
Final delivery	Thu, 3 Dec 2026	Characters, verification, and release (P3)	Final build, test results, documentation, and demo are handed over.

4 Delivery details

Delivery 1: Foundation and discovery (1 to 12 November)

- Docker/Compose environment, React client and FastAPI modular monolith skeleton, PostgreSQL schema and migrations, Redis connection, health checks, and test harness.
- Approved-content record model with source, review status, language status, owner, and update date.
- Arabic RTL and English LTR layouts with a language switch available from the public pages.
- Small reviewed bilingual content pack: one gallery or area and a few artefacts.
- Museum story and home page, map browsing, place details with nearby places, and the official visit handoff.
- Start the map-source and historical-source research on day one; if no structured map is supplied, build a small manually curated dataset from an authorized plan.

Acceptance: a visitor can search a place or artefact and see approved details with location context; missing map facts are shown as unavailable, never guessed.

Delivery 2: Grounded Concierge and history (15 to 26 November)

- Agent contracts (versioned JSON) and the Central Concierge with intent routing.
- Map Explorer and Era Historian specialists using approved content only.
- RAG with PostgreSQL and pgvector: ingestion worker, embeddings from approved content, and filtered retrieval with source IDs.
- AI Gateway with one selected provider (chosen after benchmarking two candidates on the same test set) and a deterministic static fallback.
- Safety and Provenance Guard on every generated public answer.
- Artefact and era pages with related content; stretch goal: Tour Planner and self-guided tours (P4).

Acceptance: questions route to the right specialist, every public answer is grounded, scoped, and citation-ready, and the app still works when AI is unavailable. Feature freeze after this delivery.

Final delivery: Characters, verification, and release (29 November to 3 December)

- Khufu, Ahmose I, and Cleopatra VII with explicit era boundaries and approved knowledge sets.
- Evaluation scenarios: out-of-era questions, unsupported claims, prompt injection, timeouts, outages, and Arabic/English intent parity.
- Accessibility and RTL checks on mobile and desktop; keyboard and screen-reader paths.
- Bug fixing, performance check of the public pages, and reproducible Docker images.
- Documentation: README, setup guide, architecture notes, known limitations, and the roadmap for accounts and voice.
- Demo preparation and final handover.

Acceptance: character answers stay within the selected lifetime and refuse unsupported questions safely; all release gates for content, agent, and visitor readiness are met.

5 Dependencies and risks

Dependency or risk	Impact	Mitigation
Detailed GEM map not available or reuse rights unclear	Map and nearby places cannot be built from an official source.	Request the file or permission now; fall back to a small reviewed manual dataset and show completeness status.
Approved historical content not ready	Concierge and characters cannot answer factually.	Prepare a small bilingual content pack with reviewed sources first; publish only approved records.
LLM free-tier limits or weak Arabic quality	Slow or low-quality answers during demos.	Keep providers behind the AI Gateway, benchmark two candidates, and keep the static fallback.
Short timeline	Scope may not fit.	Hold the feature freeze on 26 Nov; defer P4 to P6 first if the schedule slips.
Image rights and bilingual coverage	Pages with missing images or untranslated content.	Track rights in the source register and show translation availability explicitly.

6 Working rules

- Agree on agent JSON contracts and data ownership before parallel implementation.
- Keep provider keys server-side; no browser-to-provider calls.
- Each public content row carries source and approval metadata before it is indexed.
- Hold a short team review at each delivery date and update scope if map or source discoveries change it.
- Slip rule: if a delivery is at risk, cut scope (P4, then extra content) rather than moving the 3 December final date.